

# Jonathan Ye

Stanford, CA

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## EDUCATION

### Stanford University

B.S. in Computer Science, GPA: 3.95

Stanford, CA

Sep. 2023 – June 2027

## EXPERIENCE

### Citadel

June 2026 – Aug. 2026

Quantitative Developer Intern (Equity Quantitative Research team)

New York, NY

- Built a memory system to help Quantitative Researchers effectively retrieve credible information from past work
- Designed a RAG pipeline that pre-processes multi-modal data, provides efficient access to unstructured documents and financial metrics through MCP, and cleans/compacts curated memory context on a daily basis
- Iteratively tested alongside LLMs and ran evals against complex financial workflows
- Partnered with Quantitative Researchers to ensure the project met their needs

### Amazon

June 2025 – Sep. 2025

Software Engineer Intern

Dallas, TX

- Built a scalable, full-stack AI agent service with Python, AWS, and React.js to analyze HR case workflows and pinpoint relevant logs among millions generated per hour across CloudWatch accounts; available to internal teams
- Improved agent response time by 400% by migrating packaging from Docker to Amazon Peru, optimizing build size (1 GB → 200 MB), eliminating bottlenecks via CloudWatch log debugging, and fine-tuning the agent knowledge
- Designed and implemented security/access features - including cross-account CloudWatch log access and team-specific account and knowledge base access - using AWS CDK, IAM roles, and Harmony internal front-end
- Collaborated with team members, conducted extensive research on proprietary Amazon technologies, and iteratively refined agent design based on feedback and team-specific workflows

### Stanford Cybersecurity and AI Lab

Dec. 2024 – June 2025

Researcher

Stanford, CA

- Enhanced Cybench and Bountybench's functionality—cybersecurity benchmarking tools for LLM models—by developing and integrating high-quality CTF challenge and bug bounty tasks into the CI using Python and Docker
- Interacted with server-side software through the configuration and testing of API requests and endpoints
- Improved user productivity by 300% by adding features on front-end and agent workflow on the back-end
- Safely implemented changes by developing unit tests and collaborating with reviewers to ensure code integrity

## PROJECTS

### Poke Pocket Market | Javascript/Typescript, React.js, Next.js, Firebase, Digital Ocean

- Built and deployed a trading platform for 100+ active users who play Pokemon TCG Pocket
- Designed Firestore database and algorithms that improve operation efficiency by 500% on user and card records
- Delivered a modern user interface and user experience with Tailwind CSS and ShadCN components
- Auto-update cards by data scraping 1000 cards/second with Python and storing data in Firebase Cloud Storage
- Allowed HTTPS connection using Nginx reverse proxy to redirect traffic within Digital Ocean server

### Aspen | Discord.py, OpenAI API, LangGraph, Exa API, Python

- Built an AI-powered therapist that monitors Discord channels and reaches out to individuals in distress
- Directed the flow of conversation phases with LangGraph and OpenAI prompting
- Implemented web searching/crawling with Exa API to use intake information and design a therapy plan

### Project Launchpad | Unity, C#, FL Studio, Audacity

- Created and released a Guitar Hero/Osu-inspired rhythm game utilizing all letter keys for note interactions
- Synchronized visual cues with note interactions using Unity's particle system for enhanced gameplay
- Streamlined creating and importing levels stored in JSON files through Unity's Data Persistent Path

## TECHNICAL SKILLS

**Languages:** Python, SQL, JavaScript, TypeScript, C, C++, Java, C#, HTML, CSS

**Frameworks/Libraries:** React.js, Next.js, Apache Spark, Unity, Docker, Strands Agents

**Services/Tools:** AWS, Firebase, Git/GitHub, Linux, Nginx, CloudWatch, Lambda

**Concepts:** Full-Stack, Database Systems, Data Mining, Cloud Infrastructure, Data Structures & Algorithms